

Fake News Detection: A Comprehensive Review and a Novel Framework

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Abstract

Detecting fake news poses a significant challenge in the area of Natural Language Processing (NLP). The rapid expansion of social networking platforms has not only resulted in a vast amount of information but has also accelerated the spreading of fake news. Consequently, the impact of fake news has grown, sometimes posing a threat to public safety in the real world. This study examines various methodologies for detection of fake news, including Deep Learning (DL), Machine Learning (ML), Transfer Learning, and Hybrid approaches, based on their algorithmic mechanisms. Research papers on fake news detection from the years 2018 to 2024 were collected and analyzed. Initially, these papers were gathered, and an overview of fake news detection, along with its key features, was presented. Subsequently, the available techniques were categorized, and a comprehensive description of each method was provided. The challenges faced by classification methods were also discussed. Based on the analysis a proposed framework has been discussed in the paper.

Keywords: Natural Language Processing, Fake News Detection, Deep Learning (DL), Machine Learning(ML), Transfer Learning(TL), Random Forest (RF)

1. Introduction

The Internet has revolutionized communication and interaction through its accessibility, affordability, and speed in disseminating information. As a result, social media and online portals have become more popular for news searches and reading than traditional newspapers [29]. News has a substantial impact on molding the beliefs and viewpoints of individuals. However, fake news has become an increasingly common feature in today's political landscape [26]. Fake news is purposely forged information, which is spread either to make verifiable facts doubtful or to deceive and make false information believable [16]. To address this issue, researchers and media experts have proposed fake news detectors that utilize NLP to analyze word patterns and statistical correlations of news articles [26]. Fake news detectors have made significant strides in accurately identifying manipulated news. However, their analysis tends to be superficial, focusing on whether news articles adhere to standard journalistic styles as well as norms. Detecting fake news is a crucial yet complex problem in the area of NLP. The proliferation of social media platforms has not only made information more accessible but has also hastened the distribution of false information. Consequently, the impact of fake news has expanded, sometimes even spilling over into the endangering public safety and the physical world. Given the vast volume of online content, automated fake news detection represents a practical NLP solution that can benefit all providers of online content by streamlining the

process of identifying and preventing the spread of fake news. [27].

Consequently, fake news and deception detection on social networks present unique challenges and has become an emerging research field, with future directions in data-oriented, feature-oriented, model-oriented, and application-oriented issues [16]. One significant challenge facing organizations today is the struggle to identify efficient methods for detecting false information on the internet. Fake news is often crafted to delude users, by making it hard to differentiate from real news. Nevertheless, DL techniques have proven to be more precise in detecting fake news compared to numerous machine learning techniques [28]. DL offers several advantages over machine learning, including automated feature extraction, minimal reliance on data pre-processing, the capability to extract high-dimensional features, as well as improved accuracy. The current wide availability of data and programming frameworks has further boosted the usage and robustness of DL-based approaches [29]. Previous review papers have mostly focused on data mining and ML techniques, overlooking the potential of emerging DL-based approaches such as Attention, Generative Adversarial Networks (GANs), and Bidirectional Encoder Representations for Transformers [28]. Nevertheless, detection of fake news is a complex task that cannot be solved using NLP alone. Even human readers cannot always control the authenticity of an article just by reading it. Therefore, further fact-checking is necessary to confirm the legitimacy of news beyond doubt. Hence, a DL model that considers only the NLP aspect of the article is bound to have significant limitations in the real world [30]. The prime aim of this survey is to study different methods for identifying fake news. In this survey, the identification of fake news is predominantly reliant on DL, ML, TL, and Hybrid techniques. Additionally, this survey evaluates various research studies that utilize diverse datasets, metrics, tools, as well as categories, along with the publication year of the comprised works. Furthermore, the research papers are elucidated with their limitations.

The survey is structured as given: Section 2 presents the classification of various fake news detection methods, Section 3 delves into the literature review of these methods, Section 4 discusses the challenges encountered by these methods, in section 5 proposed framework is discussed, section 6 outlines the detailed analysis of the methods and lastly, finally concludes the survey with an overall summary.

2. Classification of fake news detection methodologies

The fake news detection approaches are alienated into Deep Learning-based methods, Machine Learning-based methods, Transfer Learning-based methods, and Hybrid methods. The

categorization of Fake News Detection Techniques is seen in Figure 1.

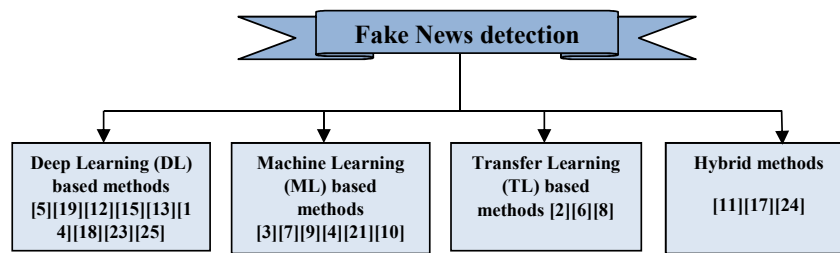


Figure 1. Categorization of Fake News Detection Methods

3. Literature Survey

The succeeding section provides a comprehensive elucidation of the collected papers, outlining the methodologies that rely on DL, ML, TL, and hybrid technologies.

A. DL-based methodologies

DL utilizes hierarchical learning or deep structured learning to process data. It employs neural networks to effectively model and address intricate problems. Below, we delve into the examination of several DL-based frameworks that have been utilized for the purpose of detecting fake news.

Saleh, H., *et al.* [5] developed an Optimized Convolutional Neural Network model (OPCNN-FAKE) for fake news detection. This network used different layers for extracting high-level as well as low-level elements. OPCNN -FAKE was also optimized by using hyper-opt optimization to select the top-performing values in each layer. This resulted in higher performance in cross-validation as well as testing. However, this technique was not used in multimodal methods with recently trained word embeddings to process visual data such as video and images. Nasir, J.A., *et al.* [12] developed a hybrid Convolutional Neural Network- Recurrent Neural Network (CNN-RNN) for fake news detection. Here, combining CNN with RNN increased the model's performance for fake news detection. The framework did not restrict other bio-inspired approaches to work on a new formulation of the issue. It included information about nodes that created and replicated news. However, it did not consider more complex neural network architectures to improve performance. Aslam, N., *et al.* [13] devised a dense deep learning model Bidirectional Long Short-Term Memory- Gated Recurrent Unit (Bi-LSTM-GRU) for Fake News Detection. Two DL models were employed in this study. The primary model, a DL-dense model, was utilized for the analysis of the nine attributes excluding statements. On the other hand, a Bi-LSTM-GRU-dense model was employed specifically for the analysis of statement attributes. By using this approach, the risk of overfitting and underfitting was minimized. However, it is important to note that this model was less interpretable and came with higher costs in terms of both time and space. Kaliyar, R.K., *et al.* [14] established a Bidirectional Encoder Representation from Transformers (BERT)-based deep learning approach (FakeBERT) for fake news detection in social media. Here, the utilization of various parallel blocks in the single-layer deep

CNN, each with distinct kernel sizes as well as filters, combined with the BERT model, proved to be effective in addressing the challenge of ambiguity in natural language understanding. This approach eliminated the need for manual feature engineering as it autonomously determined the optimal feature set. Nevertheless, a hybrid methodology was not employed for either the binary or multi-class real-world fake news databases. Amer, E., *et al.* [15] developed a Contextual-based fake news detection model based on DL. In this framework, the input data was first subjected to preprocessing, where contractions and URL were removed, along with the conversion of upper case letters to smaller case. Later, vectors were generated from the input text using the GloVe model. GloVe (Global Vectors for Word Representation) is an unsupervised learning algorithm for generating dense vector representations of words, capturing their semantic relationships based on word co-occurrence statistics from a corpus. Further, Gated Recurrent Units (GRU) and LSTM was applied for classifying the fake news. This method was more reliable and was highly effective in understanding specific domain text. Although, the method did not consider determining the authoritative information sources automatically. Mouratidis, D., *et al.* [16] devised a DL Model with Synthetic Minority Over-sampling Technique (SMOTE) and pairwise classification for fake news detection. Here, the input was integrated into the network at different levels, offering great adaptability, in order to attain the best possible classification precision. It operated without the need for a discriminator and produced top-notch data that was detailed and ideal for visual examination. However, it overlooked exploring alternative classification frameworks to detect counterfeit material. Chauhan, T. and Palivela, H., [18] developed an LSTM model for detecting misleading news. Here, vectorization or feature extraction was effectuated with the help of a tokenization process that allocated token to word embedding. Every word was represented in a vector notation by using the GloVe technique, and finally, based on the vector, LSTM classified the data. This scheme improved the detection of misleading news and this method had less computational complexity. But execution speed of this method was very low. Venkateswarlu, B., *et al.* [19] developed an Aquila Feedback Artificial tree-based Deep Residual Network (AFAT-based DRN) for detecting fake news. This approach made use of the BERT framework for converting the input to tokens. Further, the features associated with the input, like sentence level features, SentiWordNet scores, and Term Frequency-Inverse Document Frequency (TF-IDF) were mined and applied to the DRN for detecting fake news. The process was enhanced by the

application of AFAT to find the optimal weight of DRN, and the Feedback artificial tree (FAT) and Aquila optimizer (AO) were merged to obtain the AFAT. The AFAT-based DRN had minimal computational complexity and was able to capture the rich semantics in the input. However, less attention was provided towards newly occurring and time-critical events, which limited the model's performance. Ali, A.M., *et al.* [20] devised a deep ensemble model using deep learning and multilayer perceptron (SDL-MLP) for fake news detection. The method underwent construction through three distinct phases. Initially, features were derived from news content, processed through Natural Language Processing (NLP) methods, enhanced with n-gram, as well as depicted utilizing the term frequency technique. Subsequently, a deep learning-based ensemble model was developed to precisely categorize news types. Lastly, a multi-class classifier was established utilizing MLP as well as trained with features obtained from the combined outputs of DL-based binary classifiers for ultimate classification. This model identified hidden and more significant features, predicting the accuracy of news. Nevertheless, the inclusion of contextual features and extraction from other relevant sources was not considered. Roy, A., *et al.* [22] established a hybrid deep learning model composed of a Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) called DeepCnnLstm for Fake News Detection in social media. DeepCnnLstm facilitated the retrieval of spatial as well as contextual characteristics in both the forward and backward orientations of text. While this approach significantly enhanced precision in image recognition, NLP, and various other applications, it did not effectively tackle the issue of classification in multi-class, such as the potential for a news article to contain partly inaccurate information. Kaliyar, R.K., *et al.* [23] developed a Deep Neural Network (DNN) model named DeepFakeE for improving fake news detection. Here, the identification of fake information was achieved via the integration of news content and social context characteristics. Moreover, a DNN approach with fine-tuned parameters was employed to model this amalgamated representation to detect fake news. This approach demonstrated the ability to effectively learn and process extensive datasets, handle complex non-linear data relationships, and uncover concealed patterns and layers to enhance the precision of predictions. Nevertheless, it is significant to note that this framework encountered challenges such as vanishing gradients, overfitting, and a scarcity of labeled data. Kaliyar, R.K., *et al.* [25] devised EchoFakeD for improving fake news detection. Here, the dense layers in the model were configured with varying numbers of filters and dropout. A DNN was utilized, along with optimal hyper-parameters, to organize news content as well as social context-based information both separately and in combination. The model achieved a lower False Positive Rate (FPR) as well as False Negative Rate (FNR). However, it was unable to effectively tackle the challenges associated with detecting fake news using context, echo chambers, temporal, and content-based information.

B. ML-based methodologies

ML is a subset of Artificial Intelligence (AI) that permits computers to enhance their execution by learning from provided data. ML algorithms are designed to identify patterns within data sets and utilize this information to make accurate predictions.

Some of the ML-based frameworks have been utilized for fake news detection and are reviewed beneath;

Ahmad, I., *et al.* in [1] developed an ML ensemble approach along with Linguistic Inquiry and Word Count (LIWC) for Fake News Detection. The learning models underwent training and parameter tuning to achieve the highest level of accuracy. By implementing bagging and boosting, this framework successfully decreased error rates and proved to be beneficial across a diverse range of applications. Nonetheless, it did not include real-time detection of fake news in videos. Braşoveanu, A.M. and Andonie, R., [3] developed a hybrid approach that mixed ML, semantics, and NLP for Semantic Fake News Detection. The semantic fake news detection approach implemented here [3] focused on utilizing relational features, such as entities, sentiment, and facts extracted right away from the text. By extracting relational attributes and original attributes enhance the outcomes. However, the potential of leveraging these relational features in conjunction with graph neural networks for enhanced effectiveness was not taken into account. Yun, T.U. and Ahn, H., [4] devised an ML-based ensemble classification model through effective feature extraction to classify fake news. The model used an ensemble approach consisting of three known ML algorithms: Random Forest (RF), Decision Tree, and Extra Tree Classifier. By doing so, the text's complexity was decreased and unimportant features were removed from the data. In [7], Shakeel, D. and Jain, N., proposed Fast-TransE model which was utilized for the classification process, while a multi-stage naïve approach that was employed for fact validation, correction, and classification. This effective approach successfully addressed the space complexity issue. However, it failed to integrate style-based and content-based algorithms. Fayaz, M., *et al.* [9] implemented a RF classifier for Fake News Classification. Here, the efficacy of the model was determined by testing the classification system on each textual feature derived from the ISOT fake news dataset. This approach helped remove over-fitting, enhance precision, as well as lessen the training time of the predictive model. Nevertheless, the integration of deep ensemble models for accurate detection of fake news was not taken into consideration. Verma, P.K., *et al.* [10] developed a WELFake model based on Word Embedding for Fake News Detection. The final classification output was determined by this model through the utilization of a voting classifier. This classifier combined the predictions from LFS-enabled Word Embedding classification, Count Vectorizer (CV) as well as TF-IDF. By doing so, the computational complexity was reduced and the accuracy of the standard classifier was improved. However, this approach did not take into account factors such as knowledge graphs and user credibility, which could have been used for additional verification of the output produced by the WELFake model. Kaur, S., *et al.* [21] developed an automated Fake News Detection System using a Multi-level Voting Model. The integration of multiple ML models in the Multi-level Voting Model was based on their False Positive (FP) rates, resulting in a news voting classifier that provided improved prediction analysis. This model effectively addressed the trade-off between accuracy as well as efficiency, contributing potential benefits for various emerging applications such as curbing the dissemination of fake news during elections, combating managing natural calamities, and terrorism, and reducing crime

rates for the betterment of society. It had no value, however, that the detection of fake images did not utilize an annotated dataset.

C. TL-based methodologies

TL is an ML technique that involves reusing knowledge from one task to improve performance on a related task. Some of the TL-based frameworks have been utilized for fake news detection and are reviewed beneath;

Ng, K.C., *et al.* in [2] developed a domain adaptive transfer learning via an adversarial training approach for fake content detection in online platforms. Here, the combined intelligence of humans by utilizing linguistic features that are domain-invariant was utilized, extracted from a source domain that included three categories of misleading general news: biased, fake, as well as clickbait. The methodology involved utilizing datasets from different origins and assuming that text materials were classified based on predetermined domains. Nevertheless, the study did not explore the potential of ML as well as NLP techniques, like topic modeling and clustering analysis, to inspect the connection between transfer learning as well as domain specificity. Choudhry, A., *et al.* [6] established a multi-task framework, a multi-task learning approach using multi-class emotion detection for fake news and rumour detection. Here, several DL models were trained in both single-task and multi-task scenarios to conduct a thorough comparison in cross-domain environments, aiming to validate their effectiveness in enhancing generalization. Nevertheless, certain texts within the datasets resulted in inaccurate annotations. These inaccuracies caused the models to learn conflicting information, consequently resulting in decreased performance. Raza, S. and Ding, C., [8] developed a Fake News Detection model through News content and Social context (FND-NS) based on transformer architecture for fake news detection. This system successfully tackled the issue of label scarcity and was put into practice in live situations. Nevertheless, as a result of concept drift, biases emerged in the model trained on our datasets, leading to the mislabeling of certain reputable news websites. This necessitated a re-labeling and re-evaluation process utilizing up-to-date data.

D. Hybrid methodologies

Hybrid technology involves integrating various approaches to create a unified solution. This section delves into several examples of Hybrid technologies.

Albahar, M., [11] developed a Recurrent Neural Network (RNN) and Support Vector Machine (SVM) with Bidirectional GRU for fake news detection. Here, a bidirectional GRU-based RNN was employed to convert textual data, such as news content as well as comments, into numerical feature vectors. These encoded features were then utilized as input for an SVM through a radial basis function kernel to organize news as either fake or real. Notably, the comments provided additional valuable information, which significantly improved the accuracy of the detection method. However, the model's performance was compromised due to the generation of comments that lacked high quality and coherence. This issue arose from the limited diversity of highly relevant words used in the comment generation process. A hybrid linguistic and knowledge-based fake news detection system was established by Seddari, N., *et al.* [17] for Fake News Detection on social media. Here, the system merged linguistic and knowledge-based methodologies, capitalizing on their respective strengths. The system employed the holdout cross-validation technique to estimate the efficacy of the fake news detection model. Nevertheless, it overlooked the potential benefits of incorporating additional distinguishing attributes, such as style-based and visual-based features, to boost the precision of the detection system. Essa, E., *et al.* [24] developed a hybrid fake news detection system that combined BERT-based with a Light Gradient Boosting Machine (LightGBM) model fake news detection. The research entailed evaluating the efficacy of the recently devised method against four different classification strategies employing diverse word embedding methods on three genuine fake news databases. The findings demonstrated that the novel method achieved faster convergence and necessitated lower computational resources. Nonetheless, it fell short in generating synthetic networks mirroring real-world characteristics for analyzing their development.

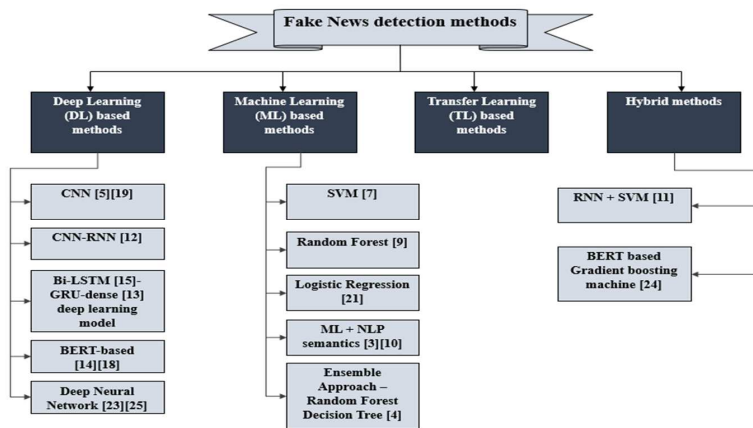


Figure 2: Summary of the Literature Review

4. Research gaps

The limitations and challenges encountered by numerous techniques of fake news detection are elucidated beneath,

Knowledge Based and Fact based Checking: Knowledge-Based checking relies on existing knowledge bases, encyclopedias, and structured datasets to verify claims against known facts. Fact-Based Checking involves verifying claims against real-time data, eyewitness accounts, or primary sources. [5] did not integrate knowledge-based and fact-based approaches to detect fake news.

Study of Echo chambers: Social media groups where users predominantly share information that aligns with their pre-existing beliefs, often leading to the reinforcement of those beliefs. [14] did not study the problem of fake news from the viewpoint of different echo chambers that exist in social media data for any social concern.

Context based Fake news detection: The same piece of news can be interpreted differently depending on the context in which it is presented, including cultural, temporal, and situational contexts. The major drawback of the Contextual-based fake news detection model in [18] was that it did not consider training ML or DL classifiers to over-extract patterns from real or fake text. The SDL-MLP developed in [20] did not consider augmenting the existing predictive learners with learners trained based on features extracted using embedding techniques for enhanced performance.

Incorporating emotion tag: Emotion tags can provide additional contextual information, as fake news and rumors often invoke specific emotional responses. The multi-task framework developed in [6] failed to incorporate fake news detection and rumor detection datasets with emotion tags to further improve the classifier's performance in detecting fake news or rumors.

Dataset Diversity: Existing fake news detection models often rely on limited datasets, which constrains their ability to generalize across different contexts and sources. The FND-NS [8] model lacks evaluation on diverse datasets like NELA-GT20 and fails to incorporate additional news sources from various websites and social media platforms. Moreover, its applicability is limited by not addressing multi-label classification, necessitating further research to enhance dataset variety and develop models capable of handling complex classification tasks in fake news detection.

5. Proposed Methodology

A novel deep learning-based technique is proposed here. The proposed method will be implemented as follows; Initially, the input data will be acquired from the dataset. The ISOT Fake News Dataset, LIAR dataset, and Kaggle Fake News Dataset are widely used for research on fake news detection. The ISOT dataset includes real and fake news articles from various sources, while the LIAR dataset offers fact-checked statements categorized by their truthfulness. The Kaggle Fake News Dataset, contains labeled news articles to help in developing models that distinguish between real and fake news. Each of these datasets provides a unique structure and set of challenges for natural language processing and machine learning tasks. Then, the acquired input data will be fed into the tokenization phase, which will be carried out using tokenization methods [31] for converting the data into

tokens. Tokenization is a crucial step in NLP that involves breaking down text into smaller units, typically words or subwords, for easier processing by algorithms. Then, the tokenized output will be subjected to feature extraction, where features, such as title, numerical sentence, Term Frequency and Inverse Document Frequency (TF-IDF) [32] based features, word count, sentence count, average word length, lin similarity score [33], sentiwordnet features [34], punctuation, emoticons, elongated words, numerical words, capitalized words, and bag of words [35] will be extracted. After feature extraction, fake news detection will be performed using the Ensemble Deep Learning model, which will be trained using hybrid Metaheuristic algorithm [36][37][38]. Hybrid metaheuristic algorithms combine elements from different metaheuristic methods, Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Jaya Algorithm (JA), CrayFish Optimization algorithm (CFO), to overcome the limitations of individual approaches and enhance the model's capability to converge towards optimal solutions. By integrating these techniques, the hybrid metaheuristic algorithm effectively explores and exploits the search space, ensuring a robust optimization of the deep learning model's hyperparameters and architecture. This approach helps in avoiding local minima and improves the generalization capability of the model. Furthermore, the adaptability of the hybrid metaheuristic algorithm ensures that it can dynamically adjust to the characteristics of the training data, leading to a more efficient training process. This adaptability is crucial in dealing with the diverse and ever-evolving nature of fake news, allowing the model to remain effective over time.

Here, the proposed framework for fake news detection will be implemented by using the Python tool. Moreover, the performance of the proposed method will be analyzed using metrics, such as accuracy, sensitivity, and specificity. Additionally, the proposed method will be evaluated for its superiority by comparing it with existing methods. Figure 3 shows the block diagram of the proposed framework for fake news detection.

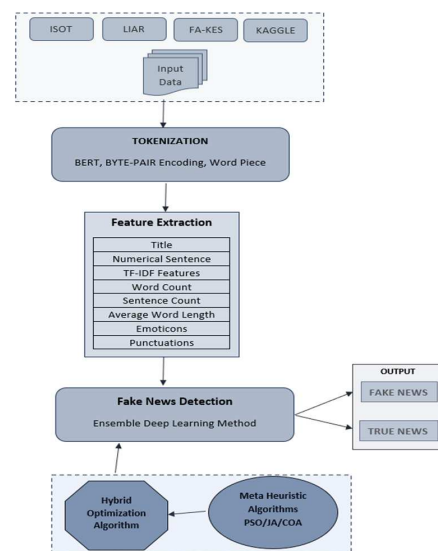


Figure 3: Proposed Methodology

6. Analysis

This section delivers an overview of the investigation and implementation of different methods for detecting fake news. Within this section, a compilation of diverse research papers is presented, accompanied by a classification of methodologies and dataset analysis.

6.1 Analysis Based on Methodologies

Fake news detection methods are primarily classified into DL-based methods, ML-based methods, TL-based methods, as well as Hybrid technologies. The breakdown of various approaches for identifying fake news is illustrated in Figure 4. Based on the analysis, it is concluded that the DL method is the most commonly used approach in precipitation forecasting.

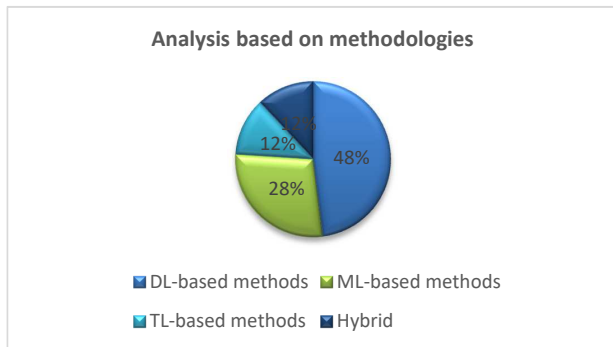


Figure 4. Evaluation based on techniques

6.2 Analysis based on Dataset

Figure 5 presents the analysis of the datasets used in the collected papers. The datasets employed for evaluating fake news detection techniques include ISOT, LIAR, News Trends, Kaggle, Reuters, FA-KES, FakeNewsNet, Fake News Corpus, PHEME, FakeNews AMT, TI-CNN, McIntire, and Buzzfeed. The figure clearly indicates that the most commonly used datasets are ISOT, LIAR, and Kaggle.

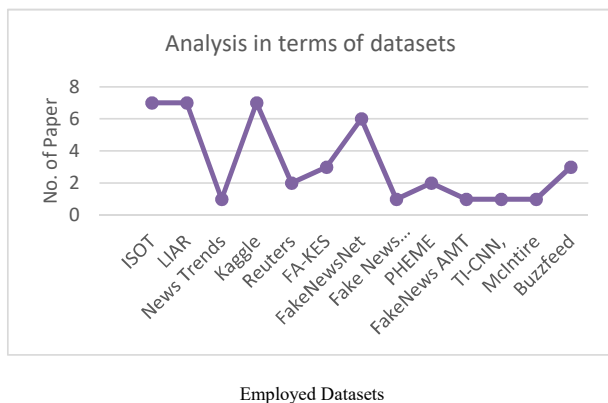


Figure 5. Evaluation based on the datasets employed

7. Conclusion and Future work

The survey compiles and explains the different approaches employed in detecting and categorizing fake news. The collected papers are classified into DL-based techniques, ML-based techniques, TL-based techniques, and Hybrid techniques. They were sourced from various publishers such as Elsevier, IEEE, Hindawi, MDPI, Research Gate, Springer, and Wiley Online Library. Each paper's literature survey is presented with a concise description, highlighting the challenges encountered. The limitations and research gaps identified provide valuable insights for further improving fake news detection. The frequently utilized databases include ISOT, LIAR, and Kaggle datasets. Moving forward, a framework using DL is proposed, and to overcome the limitation of hyper parameter tuning a hybrid metaheuristic algorithm is suggested which will be further implemented and a comparative study will be carried out. Further work will focus on developing and implementing an advanced framework for fake news detection that leverages the power of Large Language Models (LLMs) in combination with Deep Learning (DL) techniques. LLMs, such as GPT, BERT, and their variants, have shown exceptional capabilities in understanding and generating human-like text, making them ideal for enhancing the accuracy and robustness of fake news detection systems.

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